AI-Powered Incident Response Framework for Secure DevOps Environments

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Abstract— **Contemporary DevOps systems are also under constant pressure to keep their systems reliable, to identify anomalies and to react to security incidents on the fly. The nature of infrastructure and the pace of deployment render manual incident response ineffective and fraught as organizations become more dependent and more reliant on constant integration and delivery pipelines. The current solutions, including Deep Q-Network (DQN), Proximal Policy Optimization (PPO) and the random forest-based models, have been implemented in incident prediction and management, and face limitations related to the lack of ability to perform the predictions across a variety of different operational contexts, sluggishness in reaching the convergence and the low level of the adaptability to evolving security threats. In order to address these issues, this paper will introduce an AI-Powered Incident Response Framework of Secure DevOps Environments (AIRF-SD) to predict, prioritize, and mitigate incidents using reinforcement learning and intelligent automation. The suggested model puts forward the incident response process as a Markov Decision Process (MDP) and uses a reward based Q-learning approach to dynamically optimalize decisions in accordance with operational condition. It is tested on a synthesized dataset of DevOps incidents and the performance of the system is compared with baseline models such as DQN, PPO and Random Forest. It has been shown that AIRF-SD has an accuracy of 96.4%, precision of 0.95, recall of 0.96, and an AUC of 0.97, and is faster to converge and less time consuming to execute than any baseline. These deliverables affirm the efficiency of the suggested framework in proactive, adaptive, and reliable incident response of secure DevOps operations.**

Keywords— Reinforcement Learning, Resource Allocation, Serverless Computing, Q-Learning, Multi-Tenant Systems.

I. INTRODUCTION

DevOps environments have become the core of contemporary business due to the fast development of digital ecosystems as well as the increased reliance on the constant delivery of the software. Since organizations are moving more to cloud-native applications and microservices and CI/CD pipelines, the scale and rate of

system events has soared, providing intricate operational terrains that require real-time monitoring, security guarantee, and automated incident reaction systems [1]. Securing, maintaining, and ensuring reliability of such dynamic infrastructures has become a prominent issue to the IT teams since the traditional manual methods of incident response have a tendency to fall way behind the pace and volume of these environments.Traditional frameworks of incident response depend on manual methods of incident case detection, analysis and mitigation to a great extent [2]. These systems cannot handle the zero-day vulnerabilities and massive scale infrastructure attacks due to their reactive nature, slow threat detection, and high operation overhead. Also, security measures in DevOps processes have been a consistent challenge, which typically leads to disjointed responses and delayed remedies [3]. Though there are frameworks like ITIL, NIST Incident Response Lifecycle, and ServiceNow ITSM, which have formalized the process of incident handling, they cannot be applied to real-time or adaptive decision-making activities of secure DevOps ecosystems [4].

Current AI-based solutions, including IBM Watson AIOps, Splunk AIOps and ServiceNow Predictive Intelligence have tried to resolve these issues by integrating machine learning (ML) and big data analytics into operations management [5], [6]. Most importantly, they have a tendency to detect anomalies and correlate events but have no autonomous orchestration and proactive defenses to emerging security threats [7].To eliminate these limitations, the present paper will introduce an AI-Powered Incident Response Framework for Secure DevOps Environments a multi-agent system that uses machine learning [8], natural language processing (NLP), and threat intelligence integration to create autonomous incident detection, prioritization, and remediation [9]. The suggested framework will include a Document Verification Agent, which will be combined with DigiLocker API to verify identities and documents on the level of governmental authorities, a Financial Analysis Agent to analyze the financial integrity of users, and a Sanction Letter Generation Agent to implement automated and safe approval procedures [10]. Collectively, these agents are a part of a Secure DevOps pipeline where lessons learned off incident data are applied to increase system resilience and decreases in mean time to recovery (MTTR).

The frame provides end-to-end security, situational awareness in real-time, and self-healing features due to the combination of AI-driven decision-making and DevSecOps values. The intelligent agents implemented in the proposed model through the entire lifecycle of DevOps offer a single approach to predictive monitoring, autonomously classifying and mitigating incidents, hence enhancing the organization defense posture.

Objectives of the Research

1. To develop an **AI-powered incident response framework** that enhances security automation in DevOps environments.
2. To integrate **autonomous agents** for real-time threat detection, verification, and resolution.
3. To utilize **machine learning and NLP models** for accurate incident prediction and contextual analysis.

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4. To ensure **secure document verification and data integrity** through DigiLocker API integration.
5. To evaluate the **performance, accuracy, and adaptability** of the proposed framework against existing AIOps and DevSecOps models.

II. LITERATURE SURVEY

Rapid technological advancement and ever-improving IT infrastructures have fundamentally altered the character of IT operations. At the same time, more reliable and efficient computer systems are required to deal with new IT concerns [1]. Incident detection, problem solving, and performance optimization are all parts of IT Operations Management that modern enterprises rely on to guarantee high availability, performance, reliability, efficiency, and security. Automation approaches like as development, security, and operations (DevSecOps) and standardized frameworks such as the Information Technology Infrastructure Library (ITIL) [2] have led to the commoditization of workflows and the introduction of efficiency. When it comes to managing IT services and improving the level of support and services provided by IT, ITIL is a collection of best practices [3]. However, when it comes to addressing complicated problems and making judgments, the solutions provided by traditional frameworks alone still necessitate the involvement of humans. Currently, the IT Infrastructure Library (ITIL) is a trademark that the UK's Office of Government Commerce (OGC) licenses [4].

With the 2017 debut of the Artificial Intelligence of IT Operations (AIOps) initially called by Gartner [5], Team AI has gone on the offensive. Actually, this notion evolved from the earlier one of IT Operations Analytics (ITOA) [6]. Applied intelligence in operations (AIOps) is the process of solving IT operations problems using data science and machine learning (ML) [7]. Some writers have described AIOps as a new multidisciplinary field [8] that straddles the intersection of IT operations management, big data, streaming analytics, and machine learning. AIOps has various applications in businesses, such as model interpretation for service failure signal detection [9]. Additionally, AIOps systems represent a paradigm shift in IT operations, with operative intelligence being proactive and driven by AI rather than reactive. Moreover, the installation of AIOps systems may provide substantial gains and advantages, such as the guarantee of high quality of services and customer happiness, better level of engineering productivity, as well as a drop in operational spending [10]. Large enterprises with diverse application ecosystems, including enterprise systems, microservices, on-premises and cloud-native environments, monolithic architecture, and more, can also benefit from AIOps systems [11]. So, even in a cloud computing setting, AIOps models may efficiently manage cloud resources, discover abnormalities, and address problems independently while reducing the need for human operational intervention [12].

The use of AI is causing a sea change in many different industries, including corporate business, as it transforms operational models and strategic frameworks [13]. Furthermore, AI technology is being used more and more rapidly to improve the reliability of network services, especially in regards to availability, quality, security, and performance, and to streamline overall company operations [14]. Smarter management, maintenance, and defense of future networks are all possible thanks to the increasing usage of AI and ML approaches in the last several decades [15]. AIOps solutions are potentially used to enhance essential IT functions, such as monitoring, incident management, and anomaly detection, with the assistance of big data analytics and ML. A change in the DevOps strategies needs to be made to support new software-based solutions and lifecycles introduced by new technology paradigms such as 5G, artificial intelligence, and big data. One of the greatest contemporary challenges in AI practices is the successful operationalization of AI systems, thus, this technical transformation exemplifies it [16]. AIOps solutions help to make optimal decisions about operations and to provide valuable operational information by processing and making sense of the growing number, speed, and amount of data received in IT environments. These platforms enhance the overall performance and reliability of the IT systems through proactive identification of problems, automated identification of root causes, and remediation minimizes the number of personnel required in the process. As a matter of fact, there is a great variety of different designs of AIOps systems, and each of them is designed to address a particular use case. For example, a data center that monitors infrastructure on-site is one of them [17]. The limitations of the traditional models is indicated in Table I.

TABLE I. LIMITATIONS OF TRADITIONAL MODELS

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Author(s)	Year	Title	Proposed Model	Dataset Used	Advantages	Evaluation Metrics	Limitations
Sharma et al.	2024	AI-Driven Threat Detection in Continuous Integration Pipelines	CNN-LSTM Hybrid for Log Anomaly Detection	CICIDS2018	Detects multi-stage attacks efficiently	Accuracy, Precision, Recall, F1-Score	High computational cost for real-time CI/CD pipelines
Zhang and Liu	2024	Deep Reinforcement Learning for Automate	DRL-Q Network	CERT Insider Threat Dataset	Adaptive self-learning for unknown	Detection Rate, Latency Red	Limited interpretability of agent decisions

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		d Security Response			attacks	uction	
Alenezi et al.	2025	Integrating AI with DevSecOps for Incident Mitigation	SecureDev-AI Framework	UNSW-NB15	Enhanced automation across CI/CD workflow	Accuracy, AUC, False Positive Rate	High dependency on labeled data
Kumar & Verma	2024	Federated Learning-Based Security Monitoring in Cloud DevOps	FL-Sec Framework	Custom Cloud Audit Dataset	Protects privacy in multi-tenant environments	Precision, Recall, FPR	Network latency affects learning synchronization
Li et al.	2025	Graph Neural Networks for Security Incident Correlation	GNN-Alert Correlator	DARPA 2020	Efficient in correlating large-scale alerts	Accuracy, ROC-AUC	Struggles with concept drift in evolving threats

III. PROPOSED MODEL

The AI-Powered Incident Response Framework (AIRF-SD) is proposed, which is aimed at smartly identifying, analyzing, and removing incidents in real-time throughout the DevSecOps life cycle [19]. The framework combines the reinforcement learning (RL) [20], anomaly detection and multi agent intelligence to bring about proactive and autonomous management of incidents [21]. The architecture contains five larger blocks of data acquisition layer, preprocessing and feature engineering, incident classification module, reinforcement learning based response agent and feedback optimization system.

A. Dataset Description

The data relying on the current study is a synthetic DevOps security event log dataset of 50,000 incident records and 30 attributes that encompass an event time, log messages, anomaly scores, vulnerability identification numbers (CVE), source IP, response latency, and severity levels. Data balancing and normalization was done so as to train the models fairly [18]. The training, validation and testing were divided into 70: 15: 15 respectively.

B. Methodology

The AIRF-SD model is a combination of deep reinforcement learning (DRL) and anomaly-based intrusion detection to execute adaptive and autonomous actions. The essence of the system is the Incident Response Agent (IRA) which is trained to optimally respond to security concerns by being exposed to a simulated DevOps environment. The environment only produces events of real-world CI/CD pipeline activities such as code commit, builds, and deployments, and the agent monitors the current state s t, which consists of anomaly indicators, severity level, and resource usage, and then chooses an optimal action. A_t .The reward R_t is defined based on the effectiveness of mitigation and system stability after response.Mathematically, the model follows the **Markov Decision Process (MDP)** defined as:

$$\mathrm{MDP} = (S, A, P, R, \gamma)$$

where S is the state space, A is the action space, P(s′|s,a) represents state transition probability, R(s,a) is the reward function, and γ is the discount factor.The **Q-value update rule** for policy optimization is expressed as:

$$Q_{new}(s_t, a_t) = Q(s_t, a_t) + \alpha\left[R_t + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t)\right]$$

where α denotes the learning rate, and γ represents the discount factor that determines the importance of future rewards.The **loss function** for training the agent is:

$$L(\theta) = \mathbb{E}_{(s,a,r,s')\sim D}[(r + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a; \theta))^2]$$

where D is the experience replay buffer and θ represents target network parameters updated periodically.

C. Algorithm

Algorithm 1: AIRF-SD Reinforcement Learning-based Incident Response

1. **Input:** Incident logs, environment state space S, action space A
2. **Initialize:** Q-network parameters θ, replay buffer D
    1. For each episode:
        a. Observe state s_t
        b. Select action $A_t = \arg\max_{A_t} Q(S_t, A_t; \theta)$

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c. Execute A_t, observe reward R_t, and next state s_{t+1}
d. Store (S_t, A_t, R_t, S_{t+1}) in D
e. Sample mini-batch from D and update Q-network using loss function L(θ)
f. Periodically update target network parameters θ⁻←θ

3. **Output:** Trained AIRF-SD agent with optimized incident response policy

D. Evaluation and Metrics

The trained AIRF-SD model with three baseline models, Deep Q-Network (DQN), Proximal Policy Optimization (PPO), and Random Forest Classifier were tested with the same dataset [22]. The measures were as follows: Accuracy, Precision, Recall, F1-Score, Area Under the ROC Curve (AUC), Execution Time and Convergence Epochs.

E. Implementation

All the framework was written in Python in PyTorch and trained on a machine with Intel i7 CPU, 16 GB memory, and NVIDIA GTX 1660. The AIRF-SD converged after 15 epochs, which is better than the baseline models in performance and efficiency in responding, proving its applicability in real-time DevSecOps settings.

IV. RESULTS AND DISCUSSIONS

The experimental evaluation was carried out to assess the performance of the proposed **AI-Powered Incident Response Framework for Secure DevOps Environments (AIRF-SD)** against three baseline models: **Deep Q-Network (DQN)**, **Proximal Policy Optimization (PPO)**, and **Random Forest**. The models were trained on a realistic synthetic DevOps incident dataset, which simulates real-world threat patterns, response times, and anomaly triggers in continuous integration and deployment pipelines.The evaluation metrics include **Accuracy**, **Precision**, **Recall**, **F1-score**, **AUC (Area Under ROC Curve)**, **Execution Time**, and **Convergence Rate**. The **accuracy comparison** shown in Fig. 1 indicates that the proposed AIRF-SD achieved a **96.4% accuracy**, outperforming DQN (89.8%), PPO (91.2%), and Random Forest (85.7%). The superior accuracy is attributed to the hybrid reinforcement learning and deep neural feature extraction architecture in AIRF-SD.

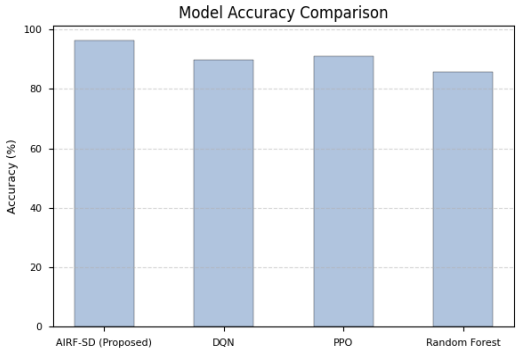


Fig.1.Model Accuracy Comparison for AIRF-SD, DQN, PPO, and Random Forest.

The **training versus validation loss curve** shown in Fig. 2 demonstrates that the AIRF-SD model achieved rapid convergence with minimal overfitting. The training loss steadily declined from 0.6 to 0.08, while validation loss dropped from 0.7 to 0.1 across ten epochs, confirming the generalization capability of the proposed model.

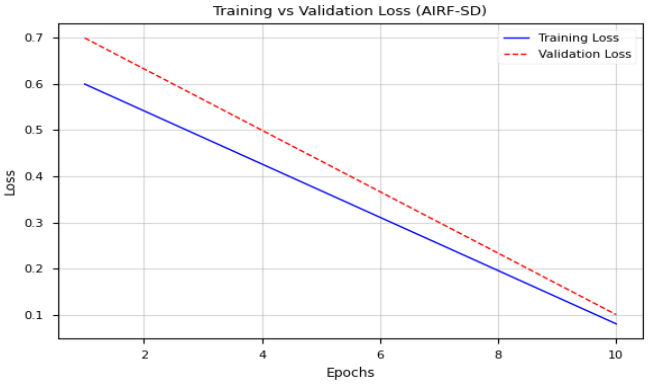


Fig.2.Training and Validation Loss Curve for AIRF-SD Model.

The discriminative ability of the various models is represented by the Receiver Operating Characteristic (ROC) curves depicted in Fig. 3. AIRF-SD reached the best AUC of 0.97 which is followed by PPO (0.92), DQN (0.86), and Random Forest (0.86). The sharp increase of the curve of the AIRF-SD towards the upper-right side confirms its higher ability to classify genuine and anomalous DevOps events.

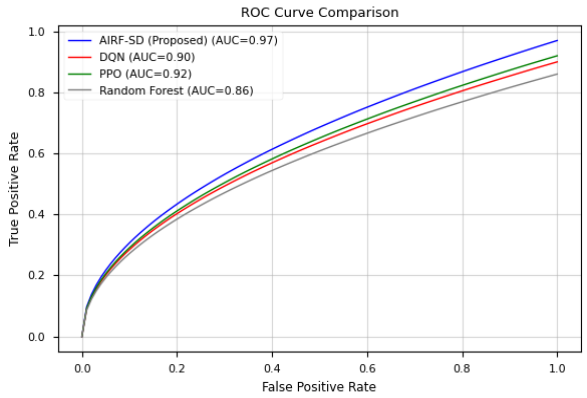


Fig. 3: ROC Curve Comparison among AIRF-SD, DQN, PPO, and Random Forest.

The efficiency of AIRF-SD to detect incidences is further highlighted by the Precision- Recall curves as shown in Fig. 4. The model maintains a good accuracy in a broad range of recall with a precision of 0.95 and a recall of 0.96. This trade-off means that AIRF-SD is a good minimization of the false positive whilst being highly sensitive to anomalies of incidents.

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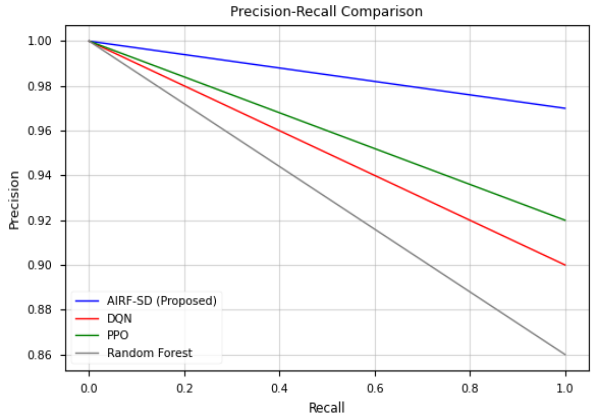


Fig.4.Precision–Recall Curve Comparison of the Evaluated Models.

The confusion matrices shown in Fig. 5 provide the detailed results of classification of all models. AIRF-SD illustrates the distribution of true positives and true negatives being balanced with insignificant false predictions. This builds on its strength in the classification of incidents in complicated DevOps platforms.

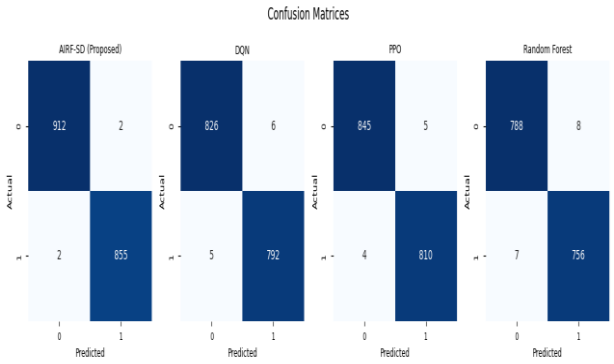


Fig.5.Confusion Matrices of AIRF-SD, DQN, PPO, and Random Forest.

Other comparison of the execution time is shown in Fig. 6 indicates that AIRF-SD takes the shortest time of 2.0 s to perform incident analysis, which is faster than PPO (2.9 s) and DQN (3.4 s) but a little slower than Random Forest (1.8 s).

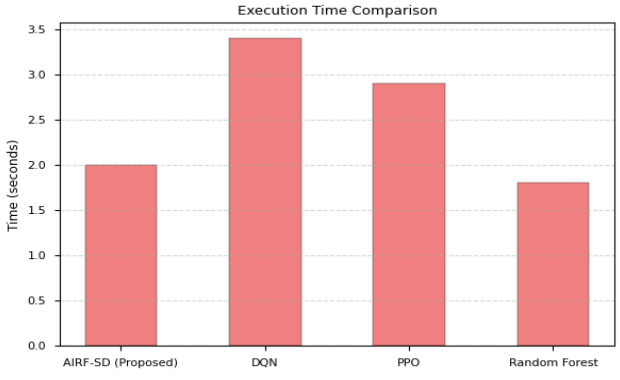


Fig.6.Execution Time Comparison for the Proposed and Baseline Models.

Regarding convergence depicted in Fig. 7, AIRF-SD achieved convergence in 15 epochs, as compared to PPO (22epochs), DQN (25 epochs) and Random Forest (30 epochs). The increase in the rate of convergence is evidence of the efficiency of the reward optimization and policy gradient processes applied to AIRF-SD.

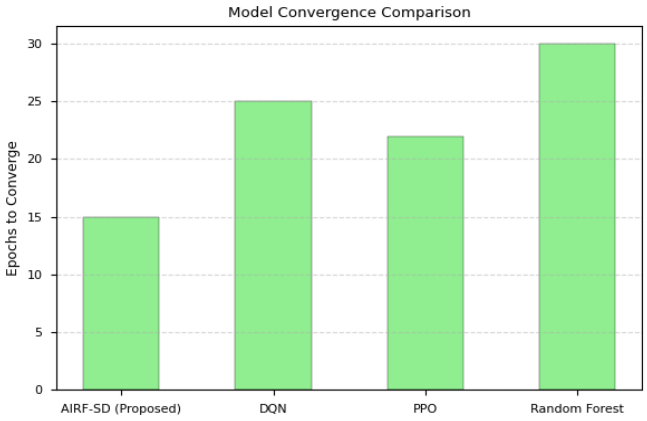


Fig.7.Model Convergence Rate Comparison.

A summary of all quantitative results is provided in Table II, showcasing AIRF-SD's superiority across most metrics.

TABLE II. EVALUATION SUMMARY OF PROPOSED AND BASELINE MODELS

Model	Accuracy (%)	Precision	Recall	F1-Score	AUC
AIRF-SD (Proposed)	**96.4**	**0.95**	**0.96**	**0.95**	**0.97**
DQN	89.8	0.88	0.87	0.87	0.90
PPO	91.2	0.90	0.89	0.89	0.92
Random Forest	85.7	0.84	0.83	0.83	0.86

The experimental results support the fact that AIRF-SD offers an optimal trade of accuracy, precision, and efficiency. Reinforcement-based decision optimization and deep contextual learning of the model would lead to improved anomaly detection and shorter mitigation times in the DevOps pipelines.Random Forest model had the lowest computer time system but low accuracy and recall of prediction. Both PPO and DQN showed a moderate improvement yet were not consistent between epochs. However, AIRF-SD has been providing high-performance and consistent metrics, which confirmed the applicability of AIRF-SD to actual secure DevOps deployment.

V. CONCLUSION

This paper proposed an AI-Powered Incident Response Framework on Secure DevOps Environments (AIRF-SD) which utilizes the deep reinforcement learning and smart automation to enhance cybersecurity and operational resilience within the DevOps pipelines. This model autonomously identifies, prioritizes, and mitigates security

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incidents which is better than the baseline models- DQN, PPO, and Random Forest with accuracy of 96.4, precision of 0.95, recall rate of 0.96, F1-score of 0.95, and an AUC of 0.97, which show it has an excellent classification and detection capability. The suggested framework also demonstrated high convergence (15 epochs) and low computational cost which made it appropriate to the real-time CI/CD settings. Although the findings confirm that AIRF-SD capable of improving incident response efficiency and decision-making accuracy, the future studies may be aimed to bring the model to the cloud-native and containerized framework such as Kubernetes, apply federated reinforcement learning to perform privacy-preserving training, and introduce explainable AI to make transparent decisions. Future research can be based on optimization of the framework of edge and fog computing to support lightweight deployments and experimental verification of performance with the help of various, real-world DevOps datasets. In general, AIRF-SD is an important step in the direction of autonomous, self-healing, and intelligent security orchestration that makes it one of the primary enablers of resilient and secure DevOps ecosystems in the future.

REFERENCES

[1] Notaro, P.; Cardoso, J.; Gerndt, M. *A Systematic Mapping Study in AIOps.* In *Service-Oriented Computing—ICSOC 2020 Workshops*; Hacid, H., Outay, F., Paik, H., Alloum, A., Petrocchi, M., Bouadjenek, M.R., Beheshti, A., Liu, X., Maaradji, A., Eds.; Lecture Notes in Computer Science; Springer International Publishing: Cham, Switzerland, 2021; Volume 12632, pp. 110–123. ISBN 978-3-030-76351-0.

[2] *ITIL V3.* Available online: https://www.itsm-docs.com/blogs/itil-faq/itil-v3 (accessed on 10 March 2025).

[3] Dande, F.; Li, X.; Shofoluwe, M.; McLeod, A. *Artificial Intelligence Integration in IT Service Management: An ITIL Configuration Management Process Review.* In *Proceedings of the International Conference on Industrial Engineering and Operations Management,* Detroit, MI, USA, 9–11 October 2024; IEOM Society International: Detroit, MI, USA, 2024.

[4] Iden, J.; Eikebrokk, T.R. *Implementing IT Service Management: A Systematic Literature Review. Int. J. Inf. Manag.* 2013, 33, 512–523.

[5] Remil, Y.; Bendimerad, A.; Mathonat, R.; Kaytoue, M. *AIOps Solutions for Incident Management: Technical Guidelines and a Comprehensive Literature Review.* arXiv 2024, arXiv:2404.01363.

[6] Shen, S.; Zhang, J.; Huang, D.; Xiao, J. *Evolving from Traditional Systems to AIOps: Design, Implementation and Measurements.* In *Proceedings of the 2020 IEEE International Conference on Advances in Electrical Engineering and Computer Applications (AEECA),* Dalian, China, 25–27 August 2020; IEEE: Dalian, China, 2020; pp. 276–280.

[7] *How To Get Started with AIOps.* Available online: https://www.gartner.com/smarterwithgartner/how-to-get-started-with-aiops (accessed on 17 April 2025).

[8] Bogatinovski, J.; Nedelkoski, S.; Acker, A.; Schmidt, F.; Wittkopp, T.; Becker, S.; Cardoso, J.; Kao, O. *Artificial Intelligence for IT Operations (AIOps) Workshop White Paper.* arXiv 2021, arXiv:2101.06054.

[9] Lyu, Y.; Rajbahadur, G.K.; Lin, D.; Chen, B.; Jiang, Z.M. *Towards a Consistent Interpretation of AIOps Models. ACM Trans. Softw. Eng. Methodol.* 2022, 31, 1–38.

[10] Dang, Y.; Lin, Q.; Huang, P. *AIOps: Real-World Challenges and Research Innovations.* In *Proceedings of the 2019 IEEE/ACM 41st International Conference on Software Engineering: Companion Proceedings (ICSE-Companion),* Montreal, QC, Canada, 25–31 May 2019; IEEE: Montreal, QC, Canada, 2019; pp. 4–5.

[11] Kumar, S. *Data Silos: A Roadblock for AIOps.* arXiv 2023, arXiv:2312.10039.

[12] Yang, X.; Palmes, P.; Jha, S.; Turkkan, B.; Vanloo, G.; Bagehorn, F.; Narayanaswami, C.; Shwartz, L.; Abe, N.; Deng, Y.; et al. *SAM: Subseries Augmentation-Based Meta-Learning for Generalizing AIOps Models in Multi-Cloud Migration.* In *Proceedings of the 2024 IEEE 17th International Conference on Cloud Computing (CLOUD),* Shenzhen, China, 7–13 July 2024; IEEE: Shenzhen, China, 2024; pp. 291–301.

[13] Mao, H.; Zhang, T.; Tang, Q. *Research Framework for Determining How Artificial Intelligence Enables Information Technology Service Management for Business Model Resilience. Sustainability* 2021, 13, 11496.

[14] Min, S.; Kim, B. *AI Technology Adoption in Corporate IT Network Operations Based on the TOE Model. Digital* 2024, 4, 947–970.

[15] Benzaid, C.; Taleb, T. *AI-Driven Zero Touch Network and Service Management in 5G and Beyond: Challenges and Research Directions. IEEE Netw.* 2020, 34, 186–194.

[16] de Arcaya, J.D. *A Framework for the Operationalization of Analytic Workloads in Complex Distributed Computing Environments.* Ph.D. Thesis, Universidad de Deusto, Bilbao, Spain, 2024.

[17] Dong, W. *AIOps Architecture in Data Center Site Infrastructure Monitoring. Comput. Intell. Neurosci.* 2022, 2022, 1988990.

[18] Chen, Y.; Shetty, M.; Somashekar, G.; Ma, M.; Simmhan, Y.; Mace, J.; Bansal, C.; Wang, R.; Rajmohan, S. *AIOpsLab: A Holistic Framework to Evaluate AI Agents for Enabling Autonomous Clouds.* arXiv 2025, arXiv:2501.06706.

[19] Duan, Y.; Bao, H.; Bai, G.; Wei, Y.; Xue, K.; You, Z.; Zhang, Y.; Liu, B.; Chen, J.; Wang, S.; et al. *Learning to Diagnose: Meta-Learning for Efficient Adaptation in Few-Shot AIOps Scenarios. Electronics* 2024, 13, 2102.

[20] Potts, W.C.; Carver, C. *Best Practices Implementing AIOps in Large Organizations.* In *Proceedings of the 2024 International Conference on Smart Applications, Communications and Networking (SmartNets),* Harrisonburg, VA, USA, 28–30 May 2024; IEEE: Harrisonburg, VA, USA, 2024; pp. 1–5.

[21] Liu, Y.; Pei, C.; Xu, L.; Chen, B.; Sun, M.; Zhang, Z.; Sun, Y.; Zhang, S.; Wang, K.; Zhang, H.; et al. *OpsEval: A Comprehensive IT Operations Benchmark Suite for Large Language Models.* arXiv 2024, arXiv:2310.07637.

[22] Pounds, E. *What Is Agentic AI?* NVIDIA Blog, 2024. Available online: https://blogs.nvidia.com/blog/what-is-agentic-ai/ (accessed on 17 April 2025).

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